CSE 242
Homework # 3

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Part 1

1. Using the support vectors, derive the maximum-margin separating line w1 x1 + w2 x2 + b = 0 (10 points), then calculate the geometric margin (10 points). Include the intercept b. Assume a linear, hard-margin SVM with no slack variables. Report the margin as the distance from the decision boundary to the closest training points; if you also report the full width between supporting boundaries, label it separately.

Ans: Using the given equation $w^1x^1 + w^2x^2 + b = 0$, we can obtain the weights, bias and geometric margin.

The given support vectors are $[(0,1),(1,1)(0,1)]$, which were obtained from finding the hyperplane of the generated plot. The associated support vectors are represented as:

$$w_1 + w_2 + b = 1$$
$$w_1 + b = -1$$
$$w_2 + b = -1$$

Since both negative support vectors are equal to each other, we can represent it as:

$$w_1 = w_2$$

We can substitute w_1 and w_2 in the first equation as $2w$, and then use the 2nd equation to find the weights:

$$2w + b = 1$$
$$b = -1 - w$$
$$2w + (-1 - w) = 1$$
$$w = 2$$

We can plug the weights into the equations to obtain the bias:

$$(2+2)+b=1=-3$$
$$(2)+b=-1=-3$$
$$(2)+b=-1=-3$$

The bias for these support vectors would be -3.

Combining the weights and bias to obtain the maximum-margin separating line would be:

$$2x_1+2x_2-3=0$$

Finally, for the geometric margin, we can use the equation:

$$\frac{1}{\sqrt{w_1^2+w_2^2}}$$

Using the obtained weights, we get:

$$\frac{1}{2^2+2^2}=0.353$$

2. In a short paragraph, explain what changes as k increases and how the results relate to the four generating groups.

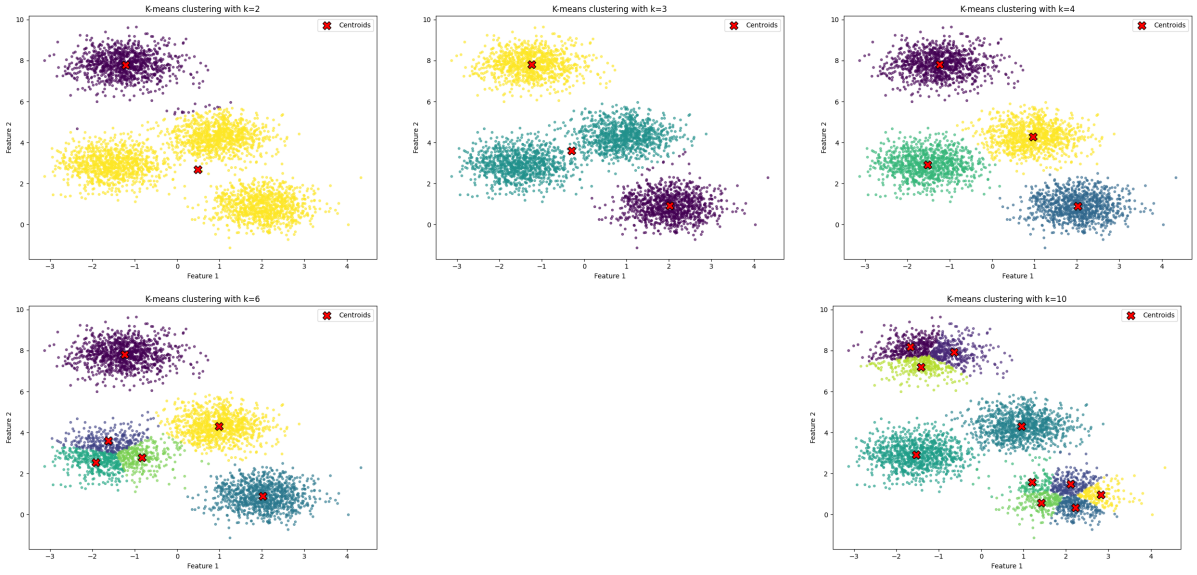


Figure 1: Clusters with incrementing centroids

Ans:

From the different results for each k, we can see that as more centroids are created, the data becomes more diverse. k_4 was able to split the data equally into its respective sections. However, when $k > 4$, it begins to over-cluster the data, causing splits between already classified clusters. Depending on whether we need to identify a subgroup within the clusters, the over-clustering could lead to misinformation.

3. Briefly explain whether K-NN repair helped and why it might help or hurt. Improvement is not guaranteed.

Ans: From the given CSV file with seed 0, it shows:

trial	baseline_accuracy	knn_accuracy
1	0.8345	0.8615
2	0.8330	0.8720
3	0.8440	0.8675
4	0.8445	0.8700
5	0.8395	0.8725
6	0.8410	0.8715
7	0.8405	0.8690
8	0.7930	0.8480
9	0.8240	0.8615
10	0.8425	0.8715
11	0.8220	0.8600
12	0.8190	0.8645
13	0.8470	0.8645
14	0.8415	0.8720
15	0.8160	0.8715
16	0.8285	0.8485
17	0.8235	0.8695
18	0.8430	0.8700
19	0.7840	0.8335
20	0.7980	0.8690

Baseline Mean: 0.8279 KNN mean: 0.8644

From this data, we can see that with the labels that were repaired, all of the trials had a minimal increase in accuracy. The five-neighbor repair method partially corrected mislabeled points before the SVM was trained. As a result, it gives the SVM a cleaner training label.